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Information and Communication Technology (ICT)





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Complete Handwritten Notes

UGC NET Paper 1

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Key Topics Covered

- Computer Fundamentals
- Internet and Networking
- E-mail, Intranet and Extranet
- Audio and Video Conferencing
- Digital Communication and Online Services
- Cybersecurity and Safe Internet Use
- ICT in Teaching and Learning
- Digital Initiatives in Higher Education
- ICT and Governance
- Emerging ICT Technologies
- Important Facts and Full Forms
- MCQs with Answers

Useful For

- UGC NET Paper 1
- Teaching Exams
- Competitive Exams
- Quick Revision
- Exam-focused MCQs

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Features

- Easy and Simple Language
- Diagrams and Tables
- Exam-oriented Content
- Handwritten Notes Style
- Complete 15 Pages

1. INTRODUCTION TO ICT

1. Meaning of ICT

- Information and Communication Technology (ICT) refers to the use of computers, internet and digital technologies to create, store, process, share and exchange information.
- It combines Information Technology (IT) and Communication Technology (CT).
- ICT helps in accessing, managing and communicating information quickly easily and efficiently.
- It plays an important role in education, research, governance and overall development of society.



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2. Components of ICT

Component	Description
Hardware	Physical devices used to input, process, store and output information.
Software	Programs and applications used to perform tasks.
Data	Raw facts and figures which are processed into useful information.
People	Users who create, use and share information.
Procedures	Rules and methods for using ICT effectively.
Network	Communication systems to connect devices.

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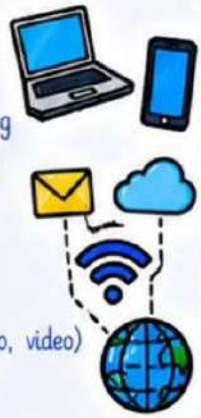
3. IT vs CT vs ICT

Aspect	Information Technology (IT)	Communication Technology (CT)	Information and Communication Technology (ICT)
Meaning	Use of computers and software to process information.	Use of communication tools to transmit information.	Combination of IT and CT for creating, storing and sharing information.
Focus	Storage, processing and management.	Transmission and exchange.	Access, management and communication.
Examples	Computers, databases, software.	Internet, mobile networks, satellite communication.	Computers, internet, e-mail, video conferencing.
Purpose	Process information.	Share information.	Create, store, process and share information.

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4. Examples of ICT Tools

- Computers and laptops
- Smartphones and tablets
- Internet and websites
- E-mail and instant messaging
- Audio and video conferencing
- Social media platforms
- Digital learning platforms (e.g., SWAYAM)
- Cloud computing tools
- Multimedia tools (text, audio, video)
- E-books and digital libraries
- Online collaboration tools (e.g., Google Workspace)



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5. Importance of ICT

- ☑ Helps in quick access to information.
- ☑ Supports teaching, learning and research.
- ☑ Improves communication and collaboration.
- ☑ Enhances efficiency and productivity.
- ☑ Provides opportunities for online and distance education.
- ☑ Supports e-governance and digital services.
- ☑ Bridges the digital divide and promotes inclusive education.
- ☑ Facilitates global connectivity and knowledge sharing.
- ☑ Useful for professional and skill development.



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6. Applications of ICT in Various Fields

Field	Applications
Education	E-learning, digital classrooms, online exams, virtual labs
Research	Online databases, e-journals, data analysis tools
Governance	E-governance, digital services, online records
Business	E-commerce, digital marketing, online banking
Healthcare	Telemedicine, electronic health records
Agriculture	Weather information, market updates, advisory services
Daily Life	Online shopping, digital payments, social media

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7. UGC NET Practice Questions (MCQs)

Q1. ICT stands for:
(A) Information and Communication Technology
(B) International Computing Technology
(C) Internet and Computer Tools
(D) Integrated Communication Technique
Ans: (A)

Q2. Which of the following is an example of communication technology (CT)?
(A) Database
(B) Satellite communication
(C) Spreadsheet
(D) Word processor
Ans: (B)

Q3. The combination of IT and CT is called:
(A) IoT
(B) ICT
(C) WWW
(D) LAN
Ans: (B)

Q4. Which of the following is NOT a component of ICT?
(A) Hardware
(B) Software
(C) People
(D) Electricity
Ans: (D)

Q5. Which of the following is a major importance of ICT in education?
(A) Reduces access to information
(B) Supports e-learning and digital resources
(C) Increases manual work
(D) Limits communication
Ans: (B)

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2. COMPUTER FUNDAMENTALS

1. What is a Computer?

- A computer is an electronic device that accepts input, processes data, stores it and produces output.
- It works according to a set of instructions called a program.
- Computers can perform complex calculations quickly and accurately.
- They are widely used in education, research, business, communication and many other fields.



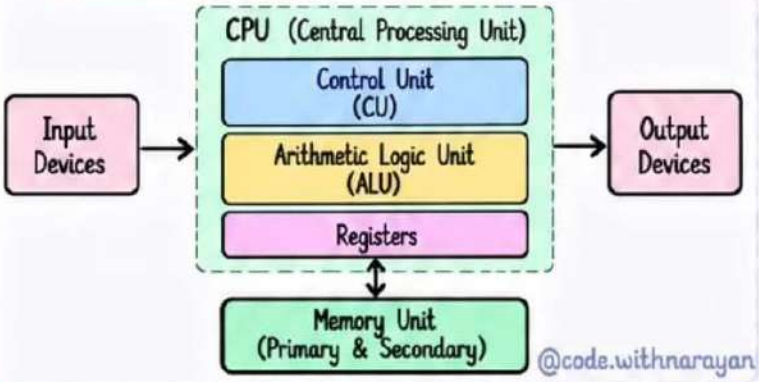
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2. Characteristics of a Computer

Speed	Can perform millions of operations per second.
Accuracy	Provides highly accurate results.
Diligence	Can work continuously without fatigue.
Storage	Can store large amount of data.
Versatility	Can perform different types of tasks.
Automation	Works automatically as per instructions.
Multitasking	Can handle multiple tasks at the same time.
Reliability	Gives consistent and reliable results.

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3. Block Diagram of a Computer



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4. Hardware vs Software

Hardware	Software
Physical components of a computer.	Programs and instructions.
Can be seen and touched.	Cannot be seen or touched.
Examples: CPU, keyboard, mouse, monitor, printer.	Examples: Operating system, MS Office, browser, antivirus.
Required for the functioning of the computer.	Tells the hardware what to do.

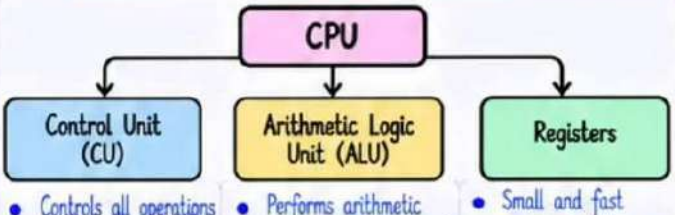
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5. Input and Output Devices

Input Devices (Give data to computer)	Output Devices (Show results)
Keyboard (text input)	Monitor (visual output)
Mouse (pointing device)	Printer (hard copy)
Microphone (audio input)	Speakers (audio output)
Scanner (image input)	Projector (large display)
Webcam (video input)	Headphones (audio output)

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6. CPU (Central Processing Unit)



- Controls all operations of the computer
- Fetches and decodes instructions
- Coordinates with other units

- Performs arithmetic operations (+, −, ×, ÷)
- Performs logical operations (AND, OR, NOT)
- Compares data

- Small and fast memory locations
- Store data and instructions temporarily
- Examples: Accumulator, Program Counter, etc.

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7. Types of Memory

Primary Memory (Main Memory)		Secondary Memory (Storage Devices)	
Type	Description	Type	Description
RAM (Random Access Memory)	Temporary memory, volatile, used to store data and programs currently in use.	Hard Disk Drive (HDD)	Magnetic storage, large capacity.
ROM (Read Only Memory)	Permanent memory, non-volatile, contains important instructions (e.g., BIOS).	Solid State Drive (SSD)	Faster than HDD, uses flash memory.
Cache Memory	Very fast memory, stores frequently used data.	CD/DVD	Optical storage devices.
		Pen Drive	Portable storage using flash memory.
		Memory Card	Used in mobile phones, cameras etc.

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8. UGC NET Practice Questions (MCQs)

Q1. Which of the following is an input device?
(A) Monitor
(B) Printer
(C) Keyboard
(D) Speaker
Ans: (C)

Q2. The brain of the computer is called:
(A) Memory
(B) Control Unit
(C) CPU
(D) ALU
Ans: (C)

Q3. Which memory is volatile in nature?
(A) ROM
(B) RAM
(C) Hard Disk
(D) CD
Ans: (B)

Q4. Which device is used to store data permanently?
(A) RAM
(B) Cache
(C) Hard Disk
(D) Register
Ans: (C)

Q5. Which of the following is an output device?
(A) Mouse
(B) Microphone
(C) Printer
(D) Scanner
Ans: (C)

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3. COMPUTER ABBREVIATIONS AND TERMINOLOGY

1. Common Computer Abbreviations

Abbreviation	Full Form
CPU	Central Processing Unit
ALU	Arithmetic Logic Unit
CU	Control Unit
RAM	Random Access Memory
ROM	Read Only Memory
BIOS	Basic Input Output System
OS	Operating System
GUI	Graphical User Interface
CLI	Command Line Interface
HDD	Hard Disk Drive
SSD	Solid State Drive
USB	Universal Serial Bus
LAN	Local Area Network
WAN	Wide Area Network
MAN	Metropolitan Area Network



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2. Internet and Web Related Terminology

Term	Meaning
URL	Uniform Resource Locator (web address)
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol Secure
WWW	World Wide Web
ISP	Internet Service Provider
DNS	Domain Name System (converts domain name to IP)
IP	Internet Protocol (unique address for a device)
FTP	File Transfer Protocol
HTML	Hyper Text Markup Language (used to create web pages)
XML	eXtensible Markup Language
PDF	Portable Document Format



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3. File and Storage Related Abbreviations

Abbreviation	Full Form
KB	Kilobyte (1024 bytes)
MB	Megabyte (1024 KB)
GB	Gigabyte (1024 MB)
TB	Terabyte (1024 GB)
CD	Compact Disc
DVD	Digital Versatile Disc
HDD	Hard Disk Drive
SSD	Solid State Drive
SD	Secure Digital (Memory Card)
PPT	PowerPoint Presentation
DOC	Document (MS Word)
XLS	Excel Spreadsheet
JPEG/JPG	Joint Photographic Experts Group (image format)
PNG	Portable Network Graphics (image format)
ZIP	Compressed file format



HDD



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4. Important ICT Terminology

Term	Meaning
Software	Set of programs and instructions.
Hardware	Physical components of a computer.
Firmware	Software stored in permanent memory.
Driver	Software that helps hardware to work.
Application	Program designed for specific tasks.
Utility	Software for maintenance and support.
Browser	Software to access and view web pages.
Search Engine	Helps to search information on the internet.
Cloud Computing	Accessing data and programs over the internet.
E-mail	Electronic mail for sending messages.
Virus	Malicious program that can harm data.
Malware	Malicious software (e.g., virus, worm, trojan).
Firewall	Security system to prevent unauthorized access.
Backup	Copy of data to prevent data loss.
Encryption	Converting data into a secure format.



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5. Operating System Examples

- An operating system (OS) manages computer hardware and software resources.
- It provides an interface between user and computer.
- Examples: Windows, Linux, macOS, Android, iOS.
- Functions: Process management, memory management, file management, device management, security.



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6. Important Units and Conversions

Unit	Abbreviation	Value
1 Byte	B	8 bits
1 Kilobyte	KB	1024 Bytes
1 Megabyte	MB	1024 KB
1 Gigabyte	GB	1024 MB
1 Terabyte	TB	1024 GB

Data Size Order

Bit < Byte < KB < MB < GB < TB

1 KB = 1024 Bytes
1 MB = 1024 KB
1 GB = 1024 MB

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7. UGC NET Practice Questions (MCQs)

Q1. What does BIOS stand for?
(A) Basic Input Output System
(B) Binary Input Output System
(C) Built-in Operating System
(D) Basic Internet Operating System
Ans: (A)

Q2. Which of the following is a web address?
(A) URL
(B) IP
(C) ISP
(D) HTTP
Ans: (A)

Q3. What is the full form of HTTP?
(A) Hyper Text Transfer Protocol
(B) High Transfer Text Protocol
(C) Hyper Transfer Tool Protocol
(D) Home Transfer Text Protocol
Ans: (A)

Q4. Which storage device uses flash memory?
(A) HDD
(B) CD
(C) SSD
(D) DVD
Ans: (C)

Q5. One Gigabyte (GB) is equal to:
(A) 1024 Bytes
(B) 1024 KB
(C) 1024 MB
(D) 1024 TB
Ans: (C)

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4. INTERNET AND WORLD WIDE WEB

1. What is Internet?

- Internet is a global network of interconnected computers and other devices.
- It allows sharing of information, communication and access to various online services.
- It works using standard protocols like TCP/IP.
- It connects millions of networks worldwide (network of networks).



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2. What is World Wide Web (WWW)?

- World Wide Web (WWW) is a collection of interlinked web pages and resources available on the Internet.
- It was invented by Tim Berners-Lee in 1989.
- It uses HTTP/HTTPS protocol to access web pages.
- WWW is a service available on the Internet.

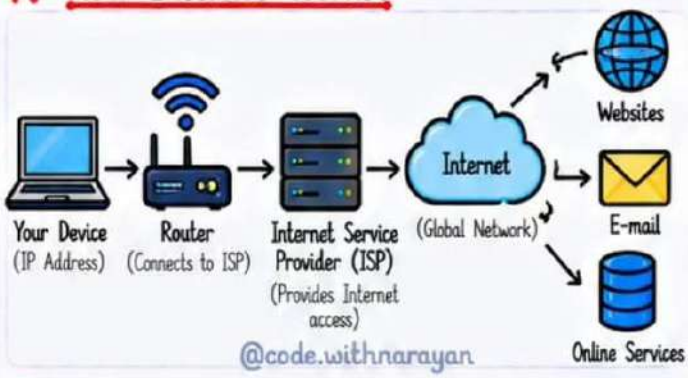


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3. Internet vs World Wide Web

Basis	Internet	World Wide Web (WWW)
Meaning	Global network of networks.	Collection of interlinked web pages.
Type	Physical network infrastructure.	A service available on Internet.
Introduced	1960s	1989 (by Tim Berners-Lee)
Purpose	Connects computers globally.	Access and share information through web pages.
Example	E-mail, FTP, chat, etc.	Websites, web pages, online documents.

4. How Internet Works?



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5. Important Components

Component	Description
IP Address	Unique address given to each device on a network.
DNS (Domain Name System)	Converts domain names (e.g., google.com) to IP addresses.
ISP (Internet Service Provider)	Provides internet access to users.
Modem	Converts digital signals to analog signals and vice versa.
Router	Connects multiple devices and routes data.
Web Browser	Software used to access web pages (e.g., Chrome, Firefox).
Search Engine	Helps to search information on the Internet (e.g., Google, Bing).

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6. Web Browsers vs Search Engines

Web Browsers	Search Engines
Software to access and view web pages.	Helps to search information on the Internet.
Examples: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera.	Examples: Google, Bing, Yahoo, DuckDuckGo.
It takes you to a website when you enter a URL.	It provides a list of websites related to your search query.

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7. HTTP vs HTTPS

Feature	HTTP	HTTPS
Full Form	Hyper Text Transfer Protocol	Hyper Text Transfer Protocol Secure
Security	Not secure (data in plain text)	Secure (data encrypted using SSL/TLS)
Port Number	80	443
Use	Used for general web browsing.	Used for secure communication (e.g., banking, e-commerce).

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8. Types of Websites

- **Static Website:** Fixed content, does not change frequently.
  Examples: Personal website, informational pages.
- **Dynamic Website:** Content changes dynamically, interactive.
  Examples: E-commerce websites, social media platforms.



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9. UGC NET Practice Questions (MCQs)

Q1. World Wide Web (WWW) was invented by:
(A) Vint Cerf
(B) Tim Berners-Lee
(C) Charles Babbage
(D) Alan Turing
Ans: (B)

Q2. Which protocol is used to access web pages on the Internet?
(A) FTP
(B) HTTP
(C) SMTP
(D) TCP
Ans: (B)

Q3. DNS stands for:
(A) Digital Network Service
(B) Domain Name System
(C) Data Network Server
(D) Direct Name Service
Ans: (B)

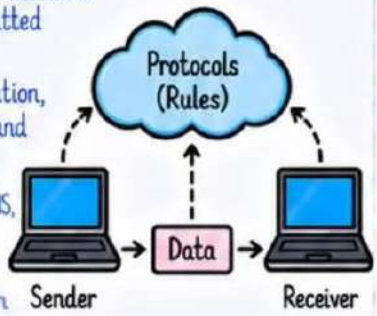
Q4. Which of the following is a web browser?
(A) Google
(B) Chrome
(C) Bing
(D) Yahoo
Ans: (B)

Q5. HTTPS is different from HTTP because:
(A) It is faster
(B) It uses encryption
(C) It requires less data
(D) It is used for e-mail
Ans: (B)

5. INTERNET PROTOCOLS AND NETWORKING

1. What are Protocols?

- A protocol is a set of rules and standards that define how data is transmitted between devices over a network.
- Protocols ensure proper communication, data formatting, error handling and security.
- Examples: TCP/IP, HTTP, HTTPS, FTP, DNS, SMTP, etc.



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2. Important Internet Protocols

Protocol	Full Form	Use / Function
TCP	Transmission Control Protocol	Reliable data transmission (split into packets and reassembled).
IP	Internet Protocol	Provides unique address to devices and routes data.
HTTP	Hyper Text Transfer Protocol	Used to access web pages (not secure).
HTTPS	Hyper Text Transfer Protocol Secure	Secure version of HTTP (uses SSL/TLS encryption).
FTP	File Transfer Protocol	Used to transfer files between devices.
SMTP	Simple Mail Transfer Protocol	Used to send e-mails.
POP3	Post Office Protocol version 3	Used to receive e-mails.
IMAP	Internet Message Access Protocol	Used to access e-mails from server.
DNS	Domain Name System	Converts domain names to IP address.

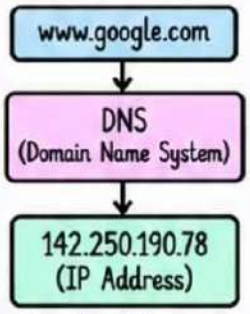
3. TCP/IP Model (4 Layers)

Layer	Functions	Examples
Application Layer	Provides network services to user applications.	HTTP, HTTPS, FTP, SMTP, DNS, IMAP, POP3
Transport Layer	Ensures reliable data transmission and error control.	TCP, UDP
Internet Layer	Handles logical addressing and routing of packets.	IP, ICMP, ARP
Network Access Layer	Manages physical transmission of data over network.	Ethernet, Wi-Fi, PPP

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4. IP Address and Domain Name

- **IP Address:** A unique numerical address assigned to each device on a network.
- Two types: IPv4 (e.g., 192.168.1.1) and IPv6 (e.g., 2001:db8::1).
- **Domain Name:** A human-readable address that is converted into an IP address using DNS.
- Example: www.google.com → 142.250.190.78



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5. Types of Networks

Type	Full Form	Coverage Area	Example
PAN	Personal Area Network	Few meters (personal)	Bluetooth, mobile hotspot
LAN	Local Area Network	Small area (building)	School, office, home
MAN	Metropolitan Area Network	City or metro area	City-wide network
WAN	Wide Area Network	Large geographical area	Internet
CAN	Campus Area Network	University or campus	College campus network
SAN	Storage Area Network	Data storage network	Enterprise storage

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6. Network Devices

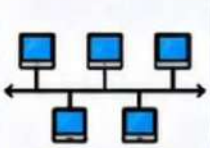
Device	Function
Router	Connects multiple networks and routes data using IP address.
Modem	Converts digital signals to analog signals and vice versa.
Switch	Connects multiple devices in a LAN and forwards data to the correct device (using MAC address).
Hub	Connects multiple devices but sends data to all.
Access Point	Provides wireless network access (Wi-Fi).
Repeater	Extends the range of a network signal.
Gateway	Connects two different networks with different protocols.



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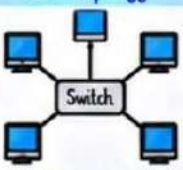
7. Network Topologies

Bus Topology



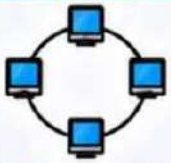
- All devices connected to a single cable.
- Simple and low cost.

Star Topology



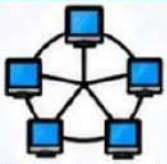
- All devices connected to a central device.
- Most commonly used.

Ring Topology



- Each device connected to two other devices.
- Data flows in one direction.

Mesh Topology



- Each device connected to many devices.
- Reliable but expensive.

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8. Important Ports

Service	Protocol	Port Number	Use
HTTP	TCP	80	Web browsing (not secure)
HTTPS	TCP	443	Secure web browsing
FTP	TCP	21	File transfer
SMTP	TCP	25	Sending e-mails
POP3	TCP	110	Receiving e-mails
IMAP	TCP	143	Accessing e-mails
DNS	UDP/TCP	53	Domain name resolution

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9. UGC NET Practice Questions (MCQs)

Q1. Which protocol is used to securely access a website?
(A) HTTP
(B) FTP
(C) HTTPS
(D) SMTP
Ans: (C)

Q2. Which layer of the TCP/IP model is responsible for routing packets?
(A) Application Layer
(B) Transport Layer
(C) Internet Layer
(D) Network Access Layer
Ans: (C)

Q3. What does DNS do?
(A) Sends e-mails
(B) Transfers files
(C) Converts domain names to IP addresses
(D) Provides web pages
Ans: (C)

Q4. Which network covers a city or metropolitan area?
(A) PAN
(B) LAN
(C) MAN
(D) WAN
Ans: (C)

Q5. Which port number is used for HTTPS?
(A) 80
(B) 21
(C) 25
(D) 443
Ans: (D)

6. INTRANET, EXTRANET AND E-MAIL

1. Internet vs Intranet vs Extranet

Feature	Internet	Intranet	Extranet
Meaning	Global network accessible to everyone.	Private network within an organization.	Restricted network that allows limited external access.
Access	Anyone	Only authorized members (internal)	Authorized internal and external users.
Purpose	Share information globally.	Internal communication and resource sharing.	Collaboration with partners, clients, suppliers.
Security	Less secure	More secure	Moderate (controlled access)
Example	www.google.com	Internal HR portal of a company	Vendor portal, client login portal

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2. E-mail: Introduction and Features



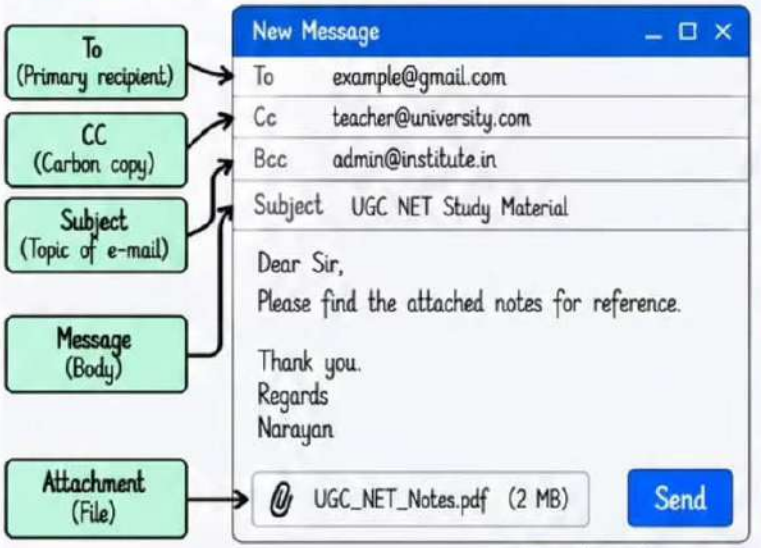
- E-mail (Electronic Mail) is a service that allows users to send and receive digital messages over the Internet.
- It is one of the most widely used communication tools in education, business and research.
- E-mails can contain text, images, documents, audio and video files as attachments.
- It is a fast, reliable, cost-effective and widely accepted mode of communication.

Popular E-mail Services

- Gmail (Google)
- Outlook (Microsoft)
- Yahoo Mail
- Rediffmail
- Zoho Mail

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3. Parts of an E-mail and Example



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4. To, CC and BCC

Field	Description
To	Primary recipient(s) of the e-mail.
CC (Carbon Copy)	Sends a copy to additional recipients (who can see other recipients).
BCC (Blind Carbon Copy)	Sends a copy to additional recipients (who cannot see other recipients).

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5. E-mail Protocols

Protocol	Full Form	Function
SMTP	Simple Mail Transfer Protocol	Used to send e-mails (from client to server).
POP3	Post Office Protocol version 3	Used to receive and download e-mails (stores on local device).
IMAP	Internet Message Access Protocol	Used to receive and manage e-mails (keeps on server, syncs across devices).

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6. E-mail Etiquette (Netiquette)

- Use a clear and meaningful subject line.
- Write in polite and professional language.
- Keep the message short and to the point.
- Avoid using all capital letters.
- Do not send unnecessary attachments.
- Check for spelling and grammar errors.
- Use CC and BCC appropriately.
- Do not share personal or sensitive information.



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7. Advantages and Limitations of E-mail

Advantages	Limitations
✅ Fast and cost-effective	❌ Risk of spam and phishing
✅ Can send files and documents	❌ Requires internet connection
✅ Useful for official communication	❌ Possibility of viruses in attachments
✅ Environment friendly (paperless)	❌ Issues of privacy and security
✅ Accessible from anywhere	❌ May lead to information overload
✅ Widely accepted and easy to use	❌ Not suitable for very urgent matters

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8. UGC NET Practice Questions (MCQs)

Q1. Which of the following is a private network within an organization?
(A) Internet
(B) Intranet
(C) Extranet
(D) WAN
Ans: (B)

Q2. Which e-mail protocol is used to send e-mails from client to server?
(A) POP3
(B) IMAP
(C) SMTP
(D) FTP
Ans: (C)

Q3. In BCC (Blind Carbon Copy), the e-mail addresses of recipients are:
(A) Visible to all
(B) Visible to some
(C) Not visible to other recipients
(D) Visible to the sender only
Ans: (C)

Q4. Which of the following is an example of an e-mail service?
(A) Google Drive
(B) Gmail
(C) Dropbox
(D) OneDrive
Ans: (B)

Q5. What is the main purpose of Extranet?
(A) Global access for everyone
(B) Internal communication only
(C) Restricted access for external partners
(D) Personal file storage
Ans: (C)

7. AUDIO AND VIDEO CONFERENCING

1. What is Audio and Video Conferencing?

- Audio and video conferencing is a real-time communication technology that allows people to interact with each other from different locations using the Internet.
- It enables sharing of audio, video, text, presentations and other digital content.
- It is widely used in education, business, research, government and personal communication.



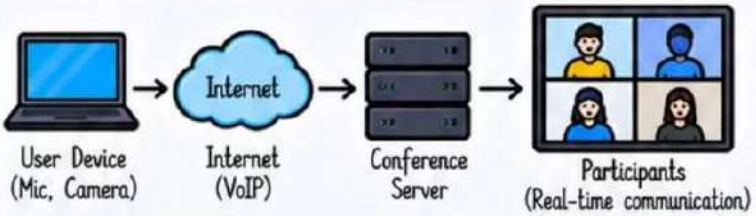
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2. Types of Conferencing

Type	Description
Audio Conferencing	Communication using only voice (audio).
Video Conferencing	Communication using both audio and video.
Web Conferencing	Includes audio, video and screen sharing through web platforms.
Teleconferencing	General term for audio/video communication over distance.
Virtual Meeting	Online meeting using internet-based tools.
Webinar	One-to-many online presentation (mainly for education and training).

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3. How Audio and Video Conferencing Works?



- Audio and video signals are converted into digital data.
- Data is transmitted over the Internet using protocols (e.g., VoIP).
- A conferencing server manages multiple participants.
- Data is received, decoded and displayed in real-time on user devices.

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4. Popular Audio/Video Conferencing Tools

Tool	Type	Key Features
Zoom	Video/Web	Meetings, webinars, screen sharing
Google Meet	Video/Web	Easy to use, integrated with Google
Microsoft Teams	Video/Web	Chat, meetings, collaboration tools
Cisco Webex	Video/Web	Secure meetings, enterprise use
Skype	Audio/Video	Individual and group calls
Jitsi Meet	Video/Web	Open source, no account required
GoToMeeting	Video/Web	Business meetings, screen sharing

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5. Important Terms

Term	Meaning
VoIP (Voice over IP)	Technology to make voice calls using the Internet.
Webinar	Online seminar with large audience (one-to-many).
Virtual Meeting	Online meeting with multiple participants.
Screen Sharing	Sharing your screen with other participants.
Chat	Text communication during a meeting.
Recording	Saving the meeting for later use.
Bandwidth	Internet speed required for smooth audio/video.



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6. Advantages and Limitations

Advantages	Limitations
✅ Real-time communication	❌ Requires good internet connection
✅ Saves time and travel cost	❌ May face technical issues
✅ Connects people globally	❌ Dependent on devices and software
✅ Supports online education	❌ Security and privacy concerns
✅ Screen sharing and collaboration	❌ Lack of face-to-face interaction
✅ Useful for business and research	❌ Can be affected by network bandwidth
✅ Environment friendly (less travel)	❌ May cause fatigue in long meetings

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7. Uses in Education and Other Fields

- **Online Classes** — Live teaching and doubt sessions
- **Virtual Meetings** — Department and faculty meetings
- **Webinars** — Training and workshops
- **Research Collaboration** — Discussion with global researchers
- **Government Services** — Online hearings and meetings
- **Business Communication** — Client meetings and team discussions

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8. UGC NET Practice Questions (MCQs)

Q1. Which technology is used to make voice calls over the Internet?
(A) VPN
(B) VoIP
(C) FTP
(D) HTTP
Ans: (B)

Q2. Which of the following is primarily a one-to-many online presentation?
(A) Virtual meeting
(B) Video conferencing
(C) Webinar
(D) Teleconferencing
Ans: (C)

Q3. Which of the following is a popular video conferencing tool developed by Google?
(A) Zoom
(B) Google Meet
(C) Cisco Webex
(D) GoToMeeting
Ans: (B)

Q4. Which feature allows a presenter to show their screen to other participants?
(A) Recording
(B) Screen sharing
(C) Chat
(D) Bandwidth
Ans: (B)

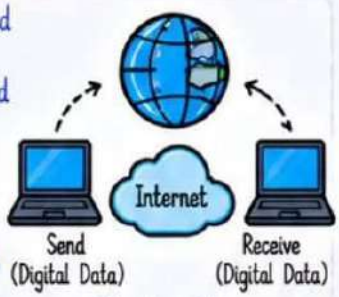
Q5. Which of the following is a limitation of audio and video conferencing?
(A) Saves travel cost
(B) Real-time communication
(C) Requires good internet connection
(D) Connects people globally
Ans: (C)

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8. DIGITAL COMMUNICATION AND ONLINE SERVICES

1. What is Digital Communication?

- Digital communication is the transmitted of data (text, audio, video, images) in digital form using electronic devices and the Internet.
- It allows fast, reliable, low-cost and global communication.
- It is widely used in education, business, government and personal communication.



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2. Types of Digital Communication

Type	Description
Text Communication	E-mails, instant messages, chat.
Audio Communication	Voice calls, VoIP, audio conferencing.
Video Communication	Video calls, video conferencing, webinars.
Multimedia Communication	Combination of text, audio, video and images.
Synchronous	Real-time communication (live chat, video call).
Asynchronous	Delayed communication (e-mail, forum posts).

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3. Social Media and Instant Messaging

Platform	Type	Use
WhatsApp	Instant Messaging	Text, voice, video, file sharing
Telegram	Instant Messaging	Secure messaging, large groups
Facebook	Social Media	Social networking, content sharing
Instagram	Social Media	Photo and video sharing
X (Twitter)	Social Media	Microblogging, news sharing
LinkedIn	Social Media	Professional networking
YouTube	Video Sharing	Educational and entertainment videos

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4. Cloud Computing and Online Collaboration

- Cloud computing provides storage, software and computing resources over the Internet.
- It allows users to access files and applications from anywhere.
- Popular cloud services: Google Drive, OneDrive, Dropbox, iCloud.
- Online collaboration tools help multiple users work on the same document in real-time.



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5. E-commerce and Online Banking

E-commerce

- Buying and selling of goods and services over the Internet.
- Examples: Amazon, Flipkart, Myntra, Meesho.
- Features: Wide selection, home delivery, digital payment, easy return policy.

Online Banking

- Performing banking transactions over the Internet.
- Services: Balance check, fund transfer, bill payment, account management.
- Examples: Net Banking, UPI, Mobile Banking Apps.

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6. E-Governance and Digital Services

- E-Governance means using digital technologies to provide government services to citizens.
- It improves transparency, efficiency and accessibility.
- Examples: Aadhaar, DigiLocker, UMANG, e-Services, Online Portals.



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7. Advantages and Challenges

Advantages

- ✅ Fast and cost-effective communication
- ✅ Global reach and easy accessibility
- ✅ Supports remote learning and work
- ✅ Wide range of online services
- ✅ Improves efficiency and productivity

Challenges

- ❌ Security and privacy concerns
- ❌ Risk of cybercrime and fraud
- ❌ Dependence on internet connectivity
- ❌ Digital divide (unequal access)
- ❌ Information overload and misinformation

8. Common Online Services

Service	Description
E-mail	Send and receive digital messages
Social Networking	Connect and share content (Facebook, Instagram)
Cloud Storage	Store and access files online (Google Drive)
E-Learning	Online courses and study material (SWAYAM, Coursera)
Online Banking	Banking transactions over the Internet
E-Commerce	Online shopping (Amazon, Flipkart)
E-Governance	Government services (Aadhaar, DigiLocker)

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9. UGC NET Practice Questions (MCQs)

Q1. Which of the following is an example of synchronous digital communication?
(A) E-mail
(B) Forum post
(C) Video conferencing
(D) Blog
Ans: (C)

Q2. Which of the following is a cloud storage service?
(A) Gmail
(B) Google Drive
(C) Paytm
(D) LinkedIn
Ans: (B)

Q3. UPI is primarily used for:
(A) Online shopping
(B) Online banking
(C) Social networking
(D) E-mail service
Ans: (B)

Q4. Which platform is mainly used for professional networking?
(A) Facebook
(B) Instagram
(C) LinkedIn
(D) YouTube
Ans: (C)

Q5. Which of the following is an e-governance service in India?
(A) Amazon
(B) DigiLocker
(C) Flipkart
(D) WhatsApp
Ans: (B)

9. CYBERSECURITY AND SAFE INTERNET USE

1. What is Cybersecurity?

- Cybersecurity means protecting computers, networks, data and digital devices from unauthorized access, attack, damage or theft.
- It helps in ensuring confidentiality, integrity and availability of information.
- It is important for individuals, organizations and governments in the digital world.



Protection from Cyber Threats

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2. Common Cyber Threats

Threat	Description	Example
Virus	Malicious program that spreads and damages files.	File infectors, macro virus
Worm	Self-replicating program that spreads over networks.	WannaCry
Trojan Horse	Malicious program disguised as a useful file.	Fake software, game crack
Spyware	Secretly collects user information.	Keyloggers
Adware	Displays unwanted advertisements.	Pop-up ads
Ransomware	Locks files and demands payment for access.	CryptoLocker
Phishing	Fraudulent attempt to steal sensitive information.	Fake e-mails, fake websites
Hacking	Unauthorized access to a system.	Password cracking

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3. Types of Malware

Type	Description
Virus	Attaches to files and spreads when file is executed.
Worm	Self-replicates and spreads over networks.
Trojan Horse	Disguised as a legitimate program.
Spyware	Tracks user activity without permission.
Adware	Shows unwanted advertisements.
Ransomware	Encrypts files and asks for a ransom.
Rootkit	Hides malicious activity and provides remote access.
Keylogger	Records keystrokes to steal sensitive information.

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4. Firewall, Antivirus and Encryption

Firewall



- Monitors and controls incoming and outgoing network traffic.
- Blocks unauthorized access.
- Can be software or hardware.

Antivirus



- Detects, prevents and removes malware.
- Scans files and programs.
- Examples: Windows Defender, Quick Heal, Norton.

Encryption



- Converts data into a coded format.
- Protects data from unauthorized access.
- Used in e-mails, online transactions and secure websites (HTTPS).

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5. Strong Password and Safe Browsing Tips

Strong Password Tips

- Use at least 8–12 characters.
- Include uppercase, lowercase, numbers and special symbols.
- Do not use personal information (e.g., name, DOB).
- Use different passwords for different accounts.
- Change password regularly.

Safe Browsing Tips

- Use trusted and secure websites (https://).
- Avoid clicking on unknown links.
- Do not share personal information online.
- Keep your browser and antivirus updated.
- Be careful with e-mail attachments.

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6. Cybersecurity Best Practices

1. Use strong and unique passwords.
2. Enable two-factor authentication (2FA).
3. Keep operating system and software updated.
4. Use antivirus and firewall.
5. Avoid suspicious links and e-mails.
6. Use secure websites (https://).
7. Do not share personal or financial information.
8. Regularly back up important data.
9. Use privacy settings on social media.
10. Be aware of phishing and online scams.



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7. Importance of Cybersecurity

- Protects personal and sensitive information.
- Prevents financial loss and data theft.
- Maintains privacy and trust.
- Ensures safe online transactions.
- Important for government, education, business and individuals.



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8. Common Cyber Attacks

Attack Type	Description
Phishing	Fake e-mails to steal personal information.
Denial of Service (DoS)	Overloads a system to make it unavailable.
Man-in-the-Middle	Intercepts communication between two parties.
SQL Injection	Exploits database vulnerabilities.
Identity Theft	Steals personal data for fraud.

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9. UGC NET Practice Questions (MCQs)

Q1. Which of the following is a malicious program that self-replicates and spreads over networks?
(A) Virus
(B) Worm
(C) Trojan Horse
(D) Adware
Ans: (B)

Q2. What does HTTPS indicate in a website address?
(A) Hyper Text Transfer Protocol
(B) Secure version of HTTP
(C) File transfer protocol
(D) E-mail protocol
Ans: (B)

Q3. Which of the following is used to detect and remove malware from a computer?
(A) Firewall
(B) Antivirus
(C) Modem
(D) Router
Ans: (B)

Q4. Phishing is an example of which type of cyber threat?
(A) Malware
(B) Social engineering
(C) Denial of Service
(D) Hacking tool
Ans: (B)

Q5. Which of the following is a good practice for safe internet use?
(A) Click on unknown links
(B) Share personal information
(C) Use strong passwords
(D) Disable antivirus
Ans: (C)

10. ICT IN TEACHING AND LEARNING

1. Meaning and Importance

- ICT in teaching and learning means the use of digital technologies to support, enhance and improve the teaching-learning process.
- It helps in creating interactive, engaging and learner-centered education.
- It provides access to a wide range of learning resources anytime and anywhere.
- It is important for quality education, skill development and lifelong learning.



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2. E-learning and Blended Learning

Type	Meaning	Features	Example
E-learning	Learning through electronic devices and internet.	Flexible, self-paced, access from anywhere.	SWAYAM, Coursera, NPTEL
Blended Learning	Combination of online and offline learning.	Mix of classroom teaching and digital learning.	School/College with LMS and offline classes
Synchronous E-learning	Real-time online interaction.	Live classes, immediate feedback.	Zoom, Google Meet
Asynchronous E-learning	Self-paced learning without real-time interaction.	Recorded lectures, flexible timing.	YouTube, NPTEL, MOOCs

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3. Learning Management System (LMS)

- LMS is a software platform used to create, deliver, manage and track learning content.
- It allows teachers to upload study material, assignments, quizzes and monitor student progress.
- It supports online and blended learning.

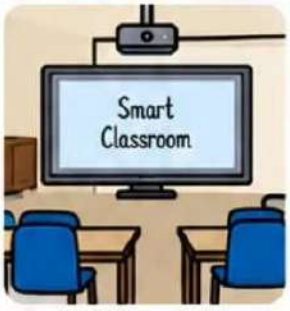
Popular LMS Platforms

Moodle (Open Source) · Google Classroom · Canvas · Microsoft Teams · Blackboard

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4. Smart Classrooms and Virtual Labs

- **Smart Classroom**: A classroom equipped with digital tools like interactive whiteboards, projectors, internet and multimedia content.
- It makes teaching more interactive and engaging.
- **Virtual Lab**: A computer-based simulation environment to perform experiments online.
- Helps students to learn practical concepts without physical equipment.



Components of Smart Classroom

Projector · Interactive Whiteboard · Computer/ Tablet · Internet Connection · Audio-Video System

Examples of Virtual Labs

NPTEL Virtual Labs · Amrita Virtual Lab · IIT Bombay Virtual Lab · OLabs (Amrita/IIT)

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5. Digital Assessment and Online Examinations

- Digital assessment uses ICT tools to evaluate student learning through online tests, quizzes and assignments.
- It provides automatic evaluation, instant feedback and better tracking.
- Online examinations are conducted over the internet using secure platforms.



Type	Description	Example
Formative Assessment	Continuous assessment during learning.	Online quizzes, assignments
Summative Assessment	Final evaluation at the end of course.	Online final exam
Objective Tests	Multiple choice, true/false questions.	MCQs, online tests
Subjective Tests	Descriptive answers.	Essay type questions
Adaptive Assessment	Questions adjust to learner's level.	AI-based testing platforms

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6. Advantages and Limitations

Advantages	Limitations
✓ Flexible and accessible learning	✗ Requires internet and digital devices
✓ Interactive and engaging content	✗ High initial setup cost
✓ Supports self-paced learning	✗ Lack of face-to-face interaction
✓ Wide range of learning resources	✗ Technical problems and maintenance
✓ Improves student motivation	✗ Need for digital skills and training
✓ Cost-effective in long term	✗ Possibility of distraction for students
✓ Helps in inclusive education	✗ Security and privacy concerns
✓ Useful during emergencies (e.g., pandemic)	✗ Not suitable for all subjects (e.g., practical/lab-based courses)

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7. UGC NET Practice Questions (MCQs)

Q1. What does LMS stand for?
(A) Learning Management System
(B) Local Management System
(C) Lecture Management Software
(D) Learning Media Service
Ans: (A)

Q2. Which of the following is an example of a MOOC platform?
(A) NPTEL
(B) Moodle
(C) Windows
(D) MS Word
Ans: (A)

Q3. In a blended learning environment, which of the following is combined?
(A) Only online learning
(B) Only offline learning
(C) Online and offline learning
(D) Virtual and physical labs
Ans: (C)

Q4. Which tool is commonly used for live online classes in education?
(A) Zoom
(B) Photoshop
(C) MS Excel
(D) WinRAR
Ans: (A)

Q5. Which of the following is a benefit of using ICT in education?
(A) Reduces access to learning resources
(B) Increases student engagement
(C) Requires no digital skills
(D) Eliminates all costs
Ans: (B)

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11. DIGITAL INITIATIVES IN HIGHER EDUCATION

1. Introduction

- Digital initiatives in higher education are programs and platforms launched by the Government of India to provide quality education using technology.
- These initiatives increase access, equity, quality and affordability of higher education.
- They support students, teachers and researchers through online courses, digital resources and virtual learning environments.



Digital India for Education

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2. Major Digital Initiatives and Platforms

Initiative	Launched by	Objective / Key Features
SWAYAM	MHRD (Now MoE)	Online courses for school, higher education and skill development.
SWAYAM PRABHA	MHRD (Now MoE)	DTH channels for 24×7 educational content.
NPTEL	IITs and IISc	High quality online courses in Engineering, Sciences and Humanities.
e-PG Pathshala	UGC	E-content for postgraduate courses in all subjects.
National Digital Library of India (NDLI)	MHRD (Now MoE) & IIT Kharagpur	One-stop digital repository of books, articles, theses and more.
DIKSHA	MoE	Platform for school education with e-learning resources.
e-ShodhSindhu	INFLIBNET (UGC)	Access to e-resources (e-journals, e-books) for higher education.
Swayam Plus	MoE	Hybrid model with credit transfer and industry-aligned courses.

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3. Details of Important Initiatives

SWAYAM

- Launched by MHRD (Now MoE).
- Provides online courses from Class 9 to Post-Graduation.
- Courses are designed by IITs, IIMs, Central and State Universities.
- Includes video lectures, reading material, quizzes and certification.

SWAYAM PRABHA

- DTH channels dedicated to education (24×7).
- Telecasts high quality educational programs for school and higher education.
- Useful for students who do not have reliable internet access.

NPTEL

- Joint initiative of IITs and IISc.
- Provides high quality video courses in Engineering, Sciences, Humanities and Management.
- Courses are free to access and certification is available through online exams.

e-PG Pathshala

- Initiated by UGC.
- Provides e-content for postgraduate courses in all subjects.
- Includes e-books, video lectures, self-assessment and quizzes.

A Gateway to All Post Graduate Courses

National Digital Library of India (NDLI)

- Developed by IIT Kharagpur under MHRD (Now MoE).
- Provides access to books, articles, theses, videos and many more learning resources.
- Useful for students, teachers and researchers.

National Digital Library of India

DIKSHA

- Launched by MoE.
- Provides e-learning content for school education.
- Offers textbooks, videos, quizzes and teacher training material.

e-ShodhSindhu

- Initiated by INFLIBNET Centre (UGC).
- Provides access to e-journals, e-books and databases.
- Supports higher education and research in India.

SWAYAM Plus

- Enhanced version of SWAYAM.
- Hybrid learning model.
- Allows credit transfer between institutions.
- Focus on industry-relevant and future-ready skills.

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4. Benefits of Digital Initiatives

- ✅ Increases access to quality education.
- ✅ Flexible and self-paced learning.
- ✅ Supports lifelong learning and skill development.
- ✅ Bridges the digital divide.
- ✅ Provides free or low-cost learning resources.
- ✅ Helps in research and academic development.

5. Challenges

- ❌ Limited internet access in rural areas.
- ❌ Digital divide and device availability.
- ❌ Lack of digital skills among users.
- ❌ Language and accessibility issues.
- ❌ Low completion rate in online courses.
- ❌ Need for continuous content update and quality assurance.

6. Impact on Higher Education

- Promotes inclusive and equitable education.
- Encourages technology-enabled teaching and learning.
- Supports multidisciplinary and skill-based learning.
- Helps in global collaboration and knowledge sharing.
- Strengthens the digital ecosystem in India.

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7. UGC NET Practice Questions (MCQs)

Q1. SWAYAM is an initiative of which ministry?
(A) UGC
(B) AICTE
(C) MHRD (Now MoE)
(D) NITI Aayog
Ans: (C)

Q2. SWAYAM PRABHA provides educational content through:
(A) Mobile App
(B) DTH Channels
(C) YouTube
(D) Radio
Ans: (B)

Q3. NPTEL is a joint initiative of:
(A) UGC and AICTE
(B) IITs and IISc
(C) MHRD and NITI Aayog
(D) NEP and NCERT
Ans: (B)

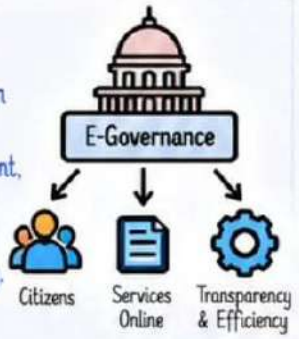
Q4. e-PG Pathshala provides e-content for:
(A) School education
(B) Undergraduate courses
(C) Postgraduate courses
(D) Research scholars only
Ans: (C)

Q5. National Digital Library of India (NDLI) is developed by:
(A) IGNOU
(B) IIT Kharagpur
(C) UGC
(D) ISRO
Ans: (B)

12. ICT AND GOVERNANCE

1. Meaning and Role of ICT in Governance

- ICT in governance means the use of digital technologies to deliver government services, share information and improve the interaction between government and citizens.
- It makes government processes more transparent, efficient, accountable and citizen-centric.
- It helps in better decision-making, reduces corruption and improves public service delivery.
- It supports good governance and inclusive development.



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2. Models of E-Governance

Model	Full Form	Meaning	Example
G2C	Government to Citizen	Delivery of services to individual citizens.	Passport Seva, Aadhaar, UMANG
G2B	Government to Business	Services to businesses and industries.	Online licensing, Company registration
G2G	Government to Government	Communication between government departments.	e-Office, NIC networks
G2E	Government to Employee	Services for government employees.	Pay slip, leave management, eHRMS

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3. Digital India Programme

- Launched in 2015 by the Government of India.
- **Aim:** To transform India into a digitally empowered society and knowledge economy.
- Focus on digital infrastructure, online services, digital literacy and inclusive growth.
- Supports e-governance, e-services, digital payments and citizen empowerment.

Digital India
Power To Empower

4. Important E-Governance Services and Platforms

Platform/Service	Launched by	Purpose	Key Features
UMANG	MeitY	One app for many government services	Access to 1000+ services (central and state)
DigiLocker	MeitY	Digital document storage	Store and access documents online
Aadhaar	UIDAI	Unique digital identity	Identity verification, DBT, e-KYC
Passport Seva	MEA	Passport related services	Online application and tracking
e-Office	MeitY (NIC)	Digital workplace for government	File management, workflow, paperless office
Bharat Bill Payment System (BBPS)	NPCI	Online bill payments	Pay electricity, water, gas, broadband bills

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5. Popular E-Governance Initiatives

- **UMANG** (One App for Many Services)
- **DigiLocker** (Digital Documents)
- **AADHAAR** (Unique Identity)
- **Passport Seva** (Passport Services)
- **e-Office** (Paperless Office)
- **GeM** (Government e-Marketplace)
- **BHIM** (UPI Payments)
- **CoWIN** (Vaccination Services)

6. Benefits and Challenges

Benefits	Challenges
✓ Faster and more efficient service delivery	✗ Digital divide (rural-urban gap)
✓ Transparency and reduced corruption	✗ Lack of digital literacy
✓ Increased citizen participation	✗ Security and privacy concerns
✓ Easy access to government services	✗ Resistance to change
✓ Cost and time saving	✗ Initial setup and maintenance cost
✓ Better decision-making with data	✗ Dependence on technology and internet connectivity
✓ Supports inclusive and sustainable development	✗ Need for continuous training and awareness

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7. Other Related Services

Service	Launched by	Purpose	Example/Use
eNAM	Ministry of Agriculture	Online trading for farmers	Agricultural produce market
Udyam Portal	Ministry of MSME	Registration for MSMEs	Udyam registration
MyGov	MeitY	Citizen participation	Share ideas, suggestions
CPGRAMS	DARPG	Grievance redressal	Lodge and track complaints
National Scholarship Portal	Ministry of Education	Scholarship services	Apply for scholarships

8. Role in Good Governance

- ICT enables transparency, accountability, responsiveness and citizen-centric governance.
- It improves the quality of public services and strengthens democratic processes.
- It helps in building a Digital India with inclusive growth and empowerment of all citizens.



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9. UGC NET Practice Questions (MCQs)

Q1. Digital India programme was launched in which year?
(A) 2010
(B) 2012
(C) 2015
(D) 2018
Ans: (C)

Q2. Which app provides access to multiple government services?
(A) DigiLocker
(B) UMANG
(C) BHIM
(D) e-Office
Ans: (B)

Q3. Aadhaar is issued by:
(A) MeitY
(B) UIDAI
(C) NPCI
(D) DARPG
Ans: (B)

Q4. Which e-governance model refers to government services to businesses?
(A) G2C
(B) G2B
(C) G2G
(D) G2E
Ans: (B)

Q5. Which platform helps in digital storage of important documents like degree certificate and driving license?
(A) UMANG
(B) DigiLocker
(C) eNAM
(D) MyGov
Ans: (B)

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13. EMERGING ICT TECHNOLOGIES

1. Introduction and Importance

- Emerging ICT technologies are new and innovative technologies that are rapidly developing and have the potential to transform various sectors.
- These technologies improve efficiency, create new opportunities and solve complex problems.
- They play an important role in education, governance, healthcare, business and daily life.
- **Examples:** Artificial Intelligence, Cloud Computing, Big Data, IoT, Blockchain, AR/VR, 5G etc.



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2. Artificial Intelligence (AI)

Aspect	Details
Meaning	AI enables machines to perform tasks that normally require human intelligence.
Key Technologies	Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Robotics.
Applications	Virtual assistants, chatbots, language translation, recommendation systems, image recognition, autonomous systems.
Use in Education	Personalized learning, intelligent tutoring systems, automated assessment.
Use in Governance	Chatbots for citizen services, data analysis, predictive decision-making, fraud detection.

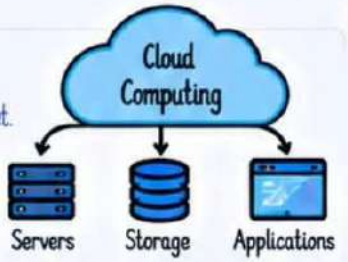


Artificial Intelligence

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3. Cloud Computing

- Cloud computing provides on-demand access to computing resources (servers, storage, software, etc.) over the Internet.
- It reduces cost, improves scalability and allows access from anywhere.
- **Popular platforms:** Google Cloud, AWS, Microsoft Azure.



Service Models

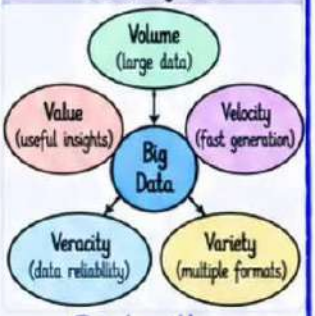
Model	Meaning
IaaS (Infrastructure as a Service)	Provides virtual servers, storage and networks.
PaaS (Platform as a Service)	Provides development platform and tools.
SaaS (Software as a Service)	Provides ready-to-use software over the Internet.

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4. Big Data and Data Analytics

- Big Data refers to very large and complex datasets that cannot be handled using traditional data processing tools.
- It is characterized by **5Vs:** Volume, Velocity, Variety, Veracity and Value.
- Data analytics helps to extract useful information and patterns from data.

5Vs of Big Data



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5. Internet of Things (IoT)

- IoT refers to a network of physical devices connected to the Internet that can collect, share and exchange data.
- These devices have sensors, software and connectivity to communicate.
- **Examples:** Smart home devices, smart wearables, smart cities, smart agriculture.



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6. Other Emerging Technologies

Technology	Meaning	Applications
Blockchain	Decentralized and secure digital ledger.	Digital payments, secure records, e-governance.
AR (Augmented Reality)	Adds digital content to the real world.	Interactive learning, training, simulations.
VR (Virtual Reality)	Creates immersive virtual environments.	Virtual labs, skill training, remote education.
5G	Next generation high-speed wireless network.	Faster connectivity, IoT, smart classrooms, smart cities.
Edge Computing	Processes data near the source of data.	Real-time processing, IoT applications.

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7. Applications in Education and Governance

In Education

- ✅ Personalized and adaptive learning
- ✅ Virtual classrooms and smart content
- ✅ AI-based assessment and feedback
- ✅ AR/VR for immersive learning
- ✅ Cloud-based LMS and digital resources
- ✅ IoT-enabled smart classrooms



Digital Education

In Governance

- ✅ AI for citizen services and chatbots
- ✅ Big Data for policy planning
- ✅ IoT for smart cities and infrastructure
- ✅ Blockchain for secure records
- ✅ Cloud for scalable e-governance services
- ✅ AR/VR for training and simulations



Smart Governance

8. Benefits and Challenges

Benefits	Challenges
✅ Improved efficiency and productivity	❌ High implementation cost
✅ Better decision-making	❌ Data privacy and security issues
✅ Enhanced learning experience	❌ Lack of digital skills
✅ Cost and time savings	❌ Technical complexity
✅ Innovation and new opportunities	❌ Infrastructure requirements
✅ Supports sustainable development	❌ Ethical and legal concerns

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9. UGC NET Practice Questions (MCQs)

Q1. Which of the following is an example of Artificial Intelligence application?
(A) Chatbot
(B) Cloud storage
(C) 5G network
(D) IoT sensor
Ans: (A)

Q2. Which service model provides ready-to-use software over the Internet?
(A) IaaS
(B) PaaS
(C) SaaS
(D) Edge Computing
Ans: (C)

Q3. Which of the following is NOT one of the 5Vs of Big Data?
(A) Volume
(B) Velocity
(C) Vision
(D) Variety
Ans: (C)

Q4. IoT devices are primarily used for:
(A) Artificial intelligence
(B) Data collection and communication
(C) Creating virtual reality
(D) High-speed internet
Ans: (B)

Q5. Which emerging technology is used to create immersive virtual environments?
(A) Augmented Reality
(B) Virtual Reality
(C) Blockchain
(D) Cloud Computing
Ans: (B)

14. IMPORTANT ICT FACTS AND QUICK REVISION

1. Important Abbreviations and Full Forms

Abbreviation	Full Form	Abbreviation	Full Form
ICT	Information and Communication Technology	IP	Internet Protocol
CPU	Central Processing Unit	DNS	Domain Name System
RAM	Random Access Memory	HTTP	Hyper Text Transfer Protocol
ROM	Read Only Memory	HTTPS	Hyper Text Transfer Protocol Secure
HDD	Hard Disk Drive	FTP	File Transfer Protocol
SSD	Solid State Drive	SMTP	Simple Mail Transfer Protocol
LAN	Local Area Network	POP3	Post Office Protocol 3
WAN	Wide Area Network	IMAP	Internet Message Access Protocol
MAN	Metropolitan Area Network	LMS	Learning Management System
PAN	Personal Area Network	MOOC	Massive Open Online Course

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2. Internet Protocols and Ports

Protocol	Port No.	Purpose
HTTP	80	Access web pages (not secure)
HTTPS	443	Secure web browsing
FTP	21	File transfer
SSH	22	Secure remote login
SMTP	25	Send e-mails
POP3	110	Receive e-mails (download)
IMAP	143	Access e-mails (sync)
DNS	53	Domain name resolution
Telnet	23	Remote login (not secure)
RDP	3389	Remote desktop access



Common Internet Protocols

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3. Computer Memory and Storage Units

Unit	Symbol	Size (in Bytes)	Equivalent
Bit	b	1 bit	0 or 1
Byte	B	8 bits	1 Byte
Kilobyte	KB	1,024 Bytes	$\approx 10^3$ Bytes
Megabyte	MB	1,024 KB	$\approx 10^6$ Bytes
Gigabyte	GB	1,024 MB	$\approx 10^9$ Bytes
Terabyte	TB	1,024 GB	$\approx 10^{12}$ Bytes
Petabyte	PB	1,024 TB	$\approx 10^{15}$ Bytes


RAM


Storage Devices

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4. Key Concepts Quick Revision

- **Internet:** Global network of computers.
- **Intranet:** Private network within an organization.
- **Extranet:** Controlled access for external users.
- **IP Address:** Unique address to identify a device.
- **DNS:** Converts domain name to IP address.
- **URL:** Address of a web page.
- **Browser:** Software to access web pages (Chrome, Firefox).
- **Search Engine:** Finds information on the internet (Google).
- **Email:** Electronic message (To, CC, BCC, Attachment).
- **Virus:** Malicious program that spreads and damages files.
- **Firewall:** Monitors and controls network traffic.
- **Cloud Computing:** On-demand access to resources.
- **IoT:** Network of physical devices connected to the internet.
- **AI:** Machines that can perform intelligent tasks.

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5. Digital Initiatives and E-Governance (Revision)

Initiative	Purpose
Digital India	Transform India into a digitally empowered society.
SWAYAM	Online courses for school and higher education.
SWAYAM PRABHA	DTH channels for educational content (24×7).
NPTEL	Online courses by IITs and IISc.
e-PG Pathshala	E-content for postgraduate courses.
NDLI	Digital repository of learning resources.
DIKSHA	E-learning platform for school education.
UMANG	One app for many government services.
DigiLocker	Digital document storage.
Aadhaar	Unique digital identity.







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6. Cybersecurity and Safe Internet Use (Revision)

Key Points	Common Threats
✅ Use strong and unique passwords.	❌ Virus
✅ Enable two-factor authentication.	❌ Worm
✅ Keep software and antivirus updated.	❌ Trojan Horse
✅ Do not click on unknown links.	❌ Spyware
✅ Avoid sharing personal information.	❌ Ransomware
✅ Use trusted and secure websites (https://).	❌ Phishing
✅ Be careful with e-mail attachments.	❌ Hacking
✅ Use firewall and antivirus.	❌ Identity Theft
✅ Be aware of phishing and online scams.	❌ Malware
	❌ Keylogger

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7. E-learning and Emerging Technologies (Revision)

E-learning Key Points

- Learning through digital devices and internet.
- LMS: Manage and deliver learning content.
- Tools: Google Classroom, Moodle, MS Teams.
- Supports online, blended and self-paced learning.
- Useful for students, teachers and professionals.

Emerging Technologies

- **AI:** Automation and intelligent systems.
- **Cloud Computing:** On-demand resources.
- **Big Data:** Analysis of large datasets.
- **IoT:** Connected devices and smart systems.
- **AR/VR:** Immersive learning experiences.
- **5G:** High-speed internet and low latency.

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8. Important Facts for Exams

- Internet was developed by ARPANET (1969).
- World Wide Web (WWW) was invented by Tim Berners-Lee (1989).
- First e-mail was sent by Ray Tomlinson (1971).
- Google was founded in 1998.
- Cloud computing popularized by Amazon (AWS).
- SWAYAM launched in 2017.
- Digital India launched in 2015.
- National Digital Library of India (NDLI) developed by IIT Kharagpur.

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9. UGC NET Practice Questions (MCQs)

Q1. Which protocol is used to transfer files over the Internet?
(A) HTTP
(B) FTP
(C) SMTP
(D) DNS
Ans: (B)

Q2. Which of the following is a cloud service model?
(A) HTTP
(B) DNS
(C) SaaS
(D) HTML
Ans: (C)

Q3. 1 GB is equal to:
(A) 1,024 KB
(B) 1,024 MB
(C) 1,024 B
(D) 1,024 TB
Ans: (B)

Q4. Which initiative provides digital learning resources for all subjects?
(A) SWAYAM
(B) DigiLocker
(C) e-Office
(D) UMANG
Ans: (A)

Q5. Which of the following helps in preventing unauthorized access to a network?
(A) Search Engine
(B) Firewall
(C) E-mail
(D) Browser
Ans: (B)

15. COMPLETE REVISION AND UGC NET PRACTICE TEST

1. Complete Syllabus Revision (Unit-wise)

1	Basics of Computers and ICT (hardware, software, types)
2	Operating Systems and Productivity Tools
3	Internet and World Wide Web (WWW)
4	Networking and Communication (LAN, WAN, etc.)
5	Web Browsers, Search Engines and Online Resources
6	Intranet, Extranet and E-mail
7	Audio and Video Conferencing
8	Digital Communication and Online Services
9	Cybersecurity and Safe Internet Use
10	ICT in Teaching and Learning (e-learning, LMS)
11	Digital Initiatives in Higher Education (SWAYAM, NPTEL, etc.)
12	ICT and Governance (e-governance, Digital India)
13	Emerging ICT Technologies (AI, Cloud, IoT, etc.)
14	Important ICT Facts and Quick Revision

Revise all topic with important points, diagrams, examples and latest updates.

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2. Important Full Forms (Quick Revision)

Abbreviation	Full Form	Abbreviation	Full Form
ICT	Information and Communication Technology	FTP	File Transfer Protocol
CPU	Central Processing Unit	SMTP	Simple Mail Transfer Protocol
RAM	Random Access Memory	POP3	Post Office Protocol 3
ROM	Read Only Memory	IMAP	Internet Message Access Protocol
HDD	Hard Disk Drive	URL	Uniform Resource Locator
SSD	Solid State Drive	LMS	Learning Management System
LAN	Local Area Network	MOOC	Massive Open Online Course
WAN	Wide Area Network	AI	Artificial Intelligence
IP	Internet Protocol	IoT	Internet of Things
DNS	Domain Name System	AR	Augmented Reality
HTTP	Hyper Text Transfer Protocol	VR	Virtual Reality
HTTPS	Hyper Text Transfer Protocol Secure	5G	Fifth Generation Mobile Network

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3. Key Concepts Summary

Topic	Key Points
Computer Basics	Input, output, processing, storage, hardware, software
Operating System	Windows, Linux, macOS, functions, types
Internet & WWW	Internet vs WWW, URL, browser, search engine
Networking	LAN, WAN, MAN, PAN, TCP/IP, IP address, DNS
E-mail	To, CC, BCC, attachments, SMTP, POP3, IMAP
Online Services	E-commerce, e-banking, e-governance, social media
Cybersecurity	Virus, malware, phishing, firewall, encryption
E-learning	LMS, MOOCs, virtual classroom, blended learning
Digital Initiatives	SWAYAM, NPTEL, NDLI, DIKSHA, UMANG, DigiLocker
Emerging Tech	AI, Cloud, Big Data, IoT, AR/VR, 5G, Blockchain

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4. Important Diagrams to Remember

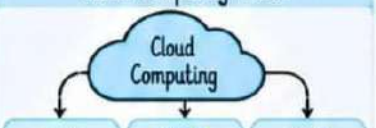
Computer System



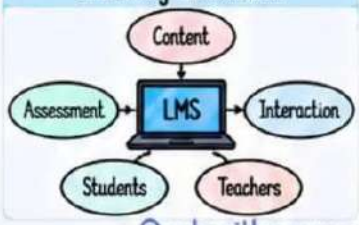
Network Types



Cloud Computing Model



E-learning Environment



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5. Digital India and E-Governance (Key Platforms)

- **Digital India** — Power To Empower
- **SWAYAM** — Online Courses
- **NPTEL** — Video Courses
- **NDLI** — Digital Library
- **DIKSHA** — E-learning
- **UMANG** — Government Services
- **DigiLocker** — Digital Documents
- **Aadhaar** — Unique Identity

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6. Cybersecurity and Safe Use (Quick Points)

- ✅ Use strong and unique passwords.
- ✅ Enable two-factor authentication.
- ✅ Keep software and antivirus updated.
- ✅ Be careful with e-mail attachments.
- ✅ Avoid clicking on suspicious links.
- ✅ Use trusted and secure websites (https://).

Common Threats

- ❌ Virus
- ❌ Worm
- ❌ Trojan Horse
- ❌ Ransomware
- ❌ Phishing
- ❌ Hacking
- ❌ Identity Theft
- ❌ Spyware



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7. Emerging Technologies (Quick Points)

- **AI:** Machine learning, deep learning, NLP, chatbots.
- **Cloud Computing:** On-demand resources, scalability.
- **Big Data:** 5Vs (Volume, Velocity, Variety, Veracity, Value).
- **IoT:** Connected devices and smart systems.
- **AR/VR:** Immersive learning experiences.
- **5G:** High-speed internet and low latency.
- **Blockchain:** Secure and decentralized records.

8. Final Revision Checklist

- ✅ All 14 units revised
- ✅ Important full forms learned
- ✅ Protocols and port numbers
- ✅ Key diagrams practiced
- ✅ Digital initiatives and platforms
- ✅ Cybersecurity topics covered
- ✅ E-learning and LMS concepts
- ✅ Emerging technologies revised
- ✅ MCQs practice completed

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9. UGC NET Practice Questions (MCQs)

Q1. Which of the following is a learning management system (LMS)?
(A) SWAYAM
(B) Moodle
(C) NPTEL
(D) Google
Ans: (B)

Q2. Which protocol is used for sending e-mails?
(A) SMTP
(B) POP3
(C) IMAP
(D) FTP
Ans: (A)

Q3. Which of the following is an example of cloud computing service model?
(A) LAN
(B) IaaS
(C) DNS
(D) HTTP
Ans: (B)

Q4. SWAYAM PRABHA provides:
(A) Online courses
(B) TV channels for educational content
(C) E-books
(D) Digital library
Ans: (B)

Q5. Which technology is used to create immersive virtual learning environments?
(A) Artificial Intelligence
(B) Internet of Things
(C) Augmented Reality/ Virtual Reality
(D) Big Data
Ans: (C)

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